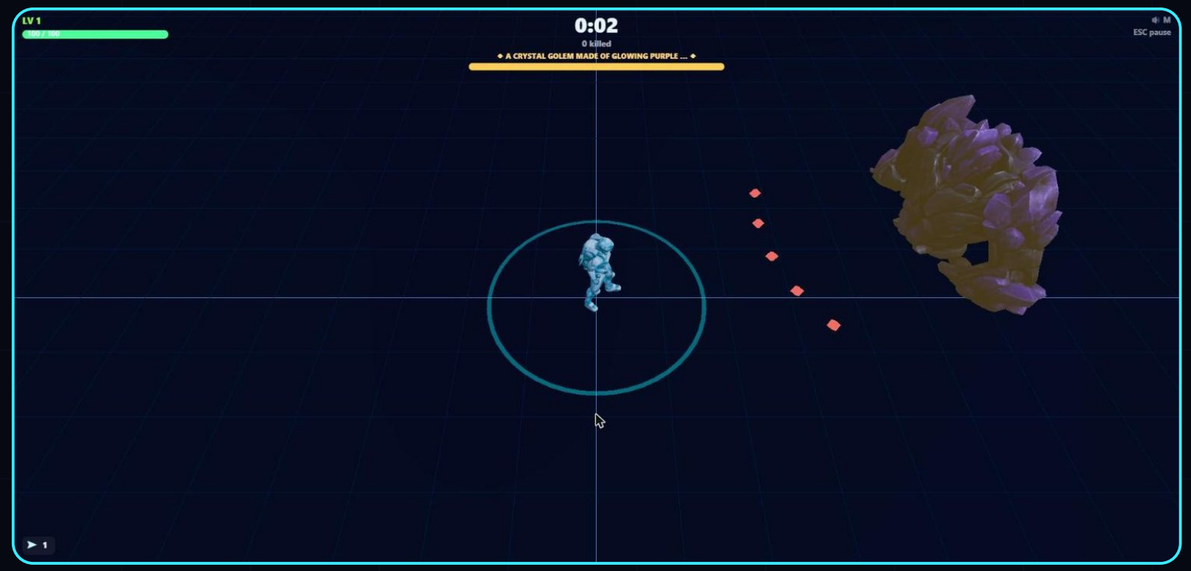
NOVA SWARM

The Generated Swarm

A browser 3D roguelite where **every character, ship and boss is generated by Aholo Lux3D**, and the next boss is whatever **you** type.

PLAY	aholo.vercel.app
SOURCE	github.com/rohanjain1648/AHOLO
BUILT BY	Rohan Jain (solo), with Claude Code as pair-engineer



A boss typed into the Forge, generated by Lux3D, arriving in a live run

Open the live link, press HANGAR, then FORGE, then PLAY SOLO. That is the whole tour.

THE IDEA

Generated 3D usually ends up in a gallery. I wanted it inside a game.

- **A game is the harshest test of generated assets.** They must be rigged, light enough for a phone, readable from a top-down camera, and fun to look at for minutes at a time.
- **A roguelite swarm is hungry for variety.** It needs many distinct, animated characters and bosses, which is exactly what a text-to-3D model can supply.
- **The twist: the player authors content.** Forge turns one typed sentence into the next boss, live, while the run is on. The prompt becomes a mechanic.

WHERE EACH JUDGING CRITERION IS ANSWERED

Use of Manycore tools	Every model, five Lux3D capabilities, and Forge at runtime
Fun and engagement	Orbit dogfight, boss intros, dash, new weapons, shatter deaths
Innovation	The prompt is a mechanic: the player authors the boss
Completeness	Live link, 200 tests, fallbacks, co-op, a Hangar of every asset

19
Lux3D models in the game

5
Lux3D capabilities used

2
game halves: surface + orbit

200
automated tests passing

THE FIVE LUX3D CAPABILITIES

- Text-to-3D, G1 (the hand-prompted cast)
- Text-to-3D, G1-Turbo (Forge + gallery bosses)
- Humanoid auto-rig
- Animation retargeting
- Part split (the two-stage capital ship)

THE GAME

What you play

SURFACE
Waves, XP, upgrades. The Warden arrives at 2:00.

→

LAUNCH
Field clears, ground falls away, camera swoops up.

→

ORBIT
Drift flight, a generated fleet, then the capital ship.

→

RE-ENTRY
Back down. The swarm is 35% tougher each loop.

Kill the Warden to launch.

Destroy the capital ship to descend.



Surface: the Warden's arrival (boss intro)



Orbit: Lux3D fighter, asteroid and capital ship

THE PROCESS

How Lux3D is used: the asset pipeline



Resumable and frugal. Every task id and result URL is saved after each step, so a crash or dropped download never pays for the same task twice. It is a dry run unless you pass **--yes**. Each model writes a log (prompt, task ids, timings, triangle counts, clips) that this report draws on.

Lux3D API	Used for
text-to-3d, G1	13 hand-prompted cast models
text-to-3d, G1-Turbo	Forge bosses, 6 gallery bosses
animations/rig	Hero, grunt, fast, tank, Warden
animations/retarget	Their Lux3D animation clips
part-split	Capital ship: bridge breaks off

Not used: image-to-3D, material transfer, articulation, World/SpatialGen, RenderCloud, LuxReal.

ONE MODEL, START TO FINISH: THE GRUNT

22 MB > 367 KB

12.7k > 5.5k triangles

- Generated in 4 min 44 s, rigged in 56 s, retargeted in 11 s.
- All 13 cast models together weigh about 3.5 MB.
- Every model keeps a log of how it was made.

PROCESS

Iteration 1: the rig that was rejected

ATTEMPT 1

Prompt: *"a rusty scrap-metal robot soldier ... **hunched posture** ..."*

Generation succeeded. The auto-rig did not:

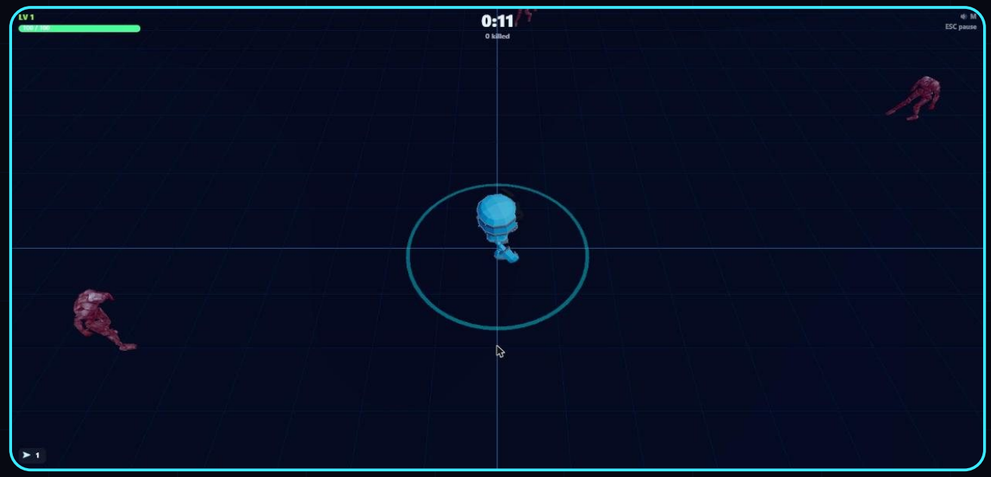
MODEL_NOT_RIGGABLE: a humanoid robot, but its pose is not a valid base stance. The body is not upright and the knees are bent.

A rejected rig did not appear to cost credits, so the lesson was cheap.

ATTEMPT 2: THE FIX

Every humanoid prompt now asks for **"standing perfectly upright with straight legs, in a T-pose"**, with a comment in cast.mjs explaining why. The pipeline also surfaces Lux3D's real error text instead of a bare status code.

Result: rigged in 56 s, four Lux3D clips in 11 s (Idle, Zombie Walk, Sprint, Death). The other four humanoids rigged first try.



The grunt in the arena, walking on a Lux3D animation



The same model in the Hangar, with its prompt and clips

PROCESS

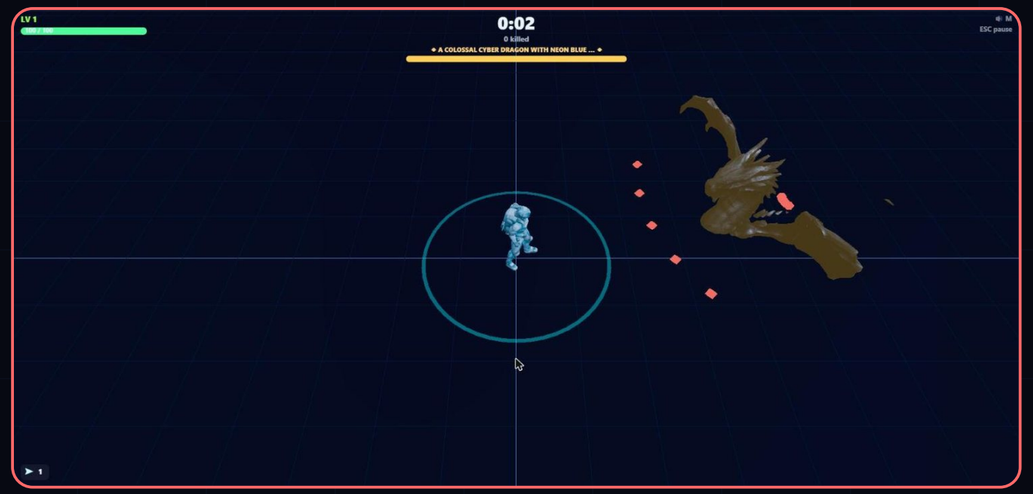
Iteration 2: making generated assets behave



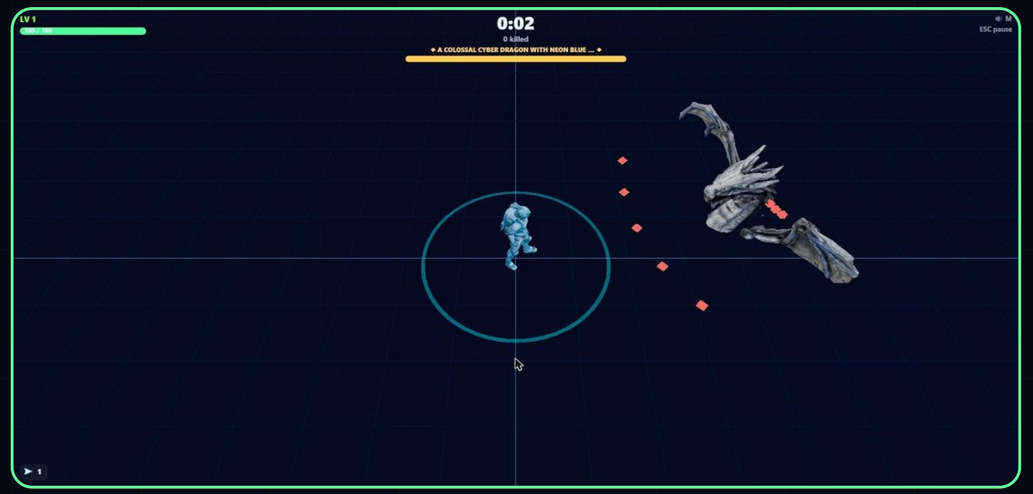
Before: ships built nose-up, the capital ship sideways



After: every nose follows the orange game-forward arrow



Before: G1-Turbo output is dark; the Warden's gold tint made it brown



After: lit by its own texture, so it reads

PROCESS

Iteration 3: from stand-in shapes to Lux3D ships



First playable orbit: low-poly placeholders I wrote in code



Same moment with the Lux3D fighter, asteroid and capital ship

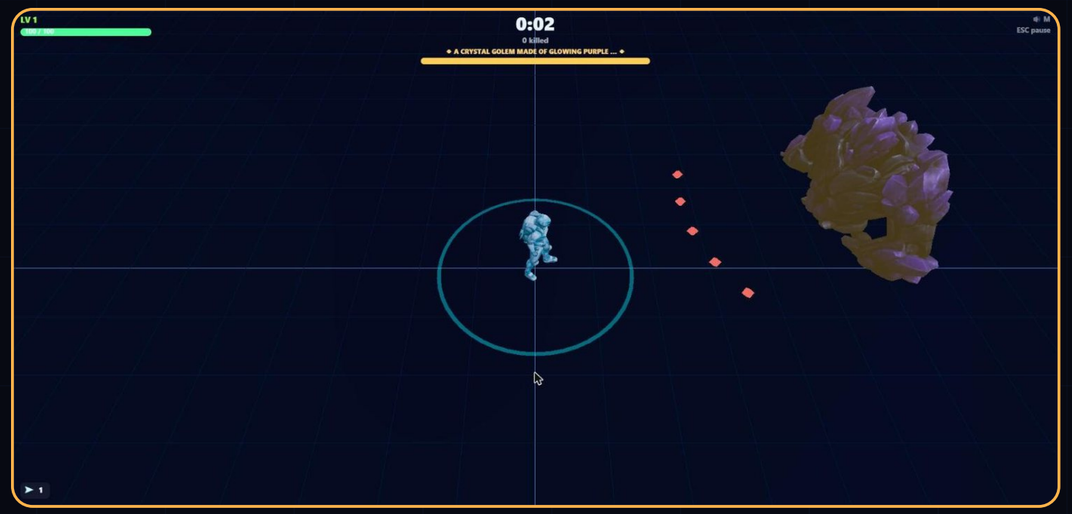
- I built the whole surface-to-orbit loop first with placeholder shapes, so the game design could be judged before spending credits.
- Then six Lux3D prompts replaced them in one batch (about 5 min each, three at a time). The game picks them up from the manifest, no code change.
- A built-in preview tool draws every model from the game camera's angle, which is how the orientation problems on the previous page were found.

THE FEATURE

Forge: you author the boss



1. Type a boss



2. About 2 to 4 minutes later it arrives as the boss

HOW IT STAYS SAFE

- The Lux3D key lives **only on the relay server**.
- Prompts are cleaned and length-limited.
- **3 forges per hour per address**, 40 per day.
- 6 pre-forged bosses as fallback: relay offline, cap hit, failed job, or credits gone.
- In co-op the host shares its boss so everyone fights the same creature.

3. IF LIVE GENERATION CANNOT RUN

The game says so, plainly, and summons the closest pre-forged Lux3D boss to what you typed:

"Live Forge is offline. Pre-forged boss: 'a colossal cyber dragon with neon blue circuit scales'"

Why it matters: credits expire before judging ends, so Forge degrades to the gallery instead of breaking, and says so.

THE FEATURE

Part split: a two-stage capital ship



Stage 1: the command bridge stands on the hull

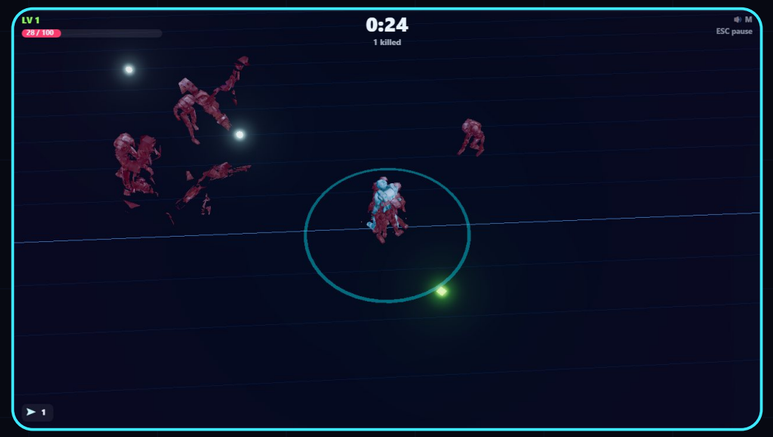


Stage 2: the bridge tore off, leaving a scorched gap

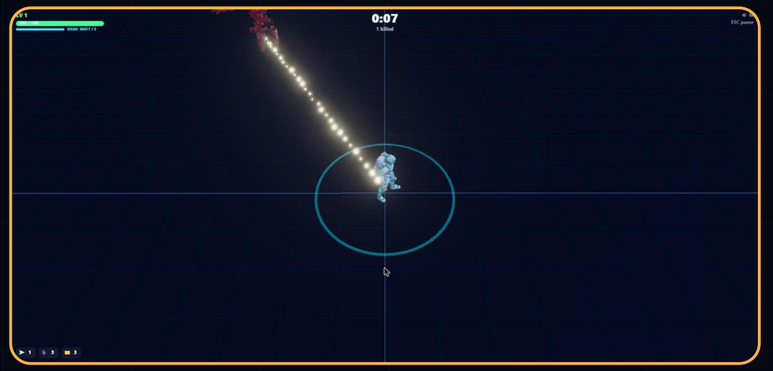
- **Lux3D part split** (94 s) separated the generated capital ship into hull and bridge. It found two parts, not a row of turrets, so the design follows what it actually gave.
- At **half health** the bridge is flung away and tumbles, BRIDGE DESTROYED appears with a screen shake, and the ship fires about **50% faster**.
- The stage rides in the network snapshot, so co-op players see the same moment. A helper script names the parts and installs the model.

THE FEATURE

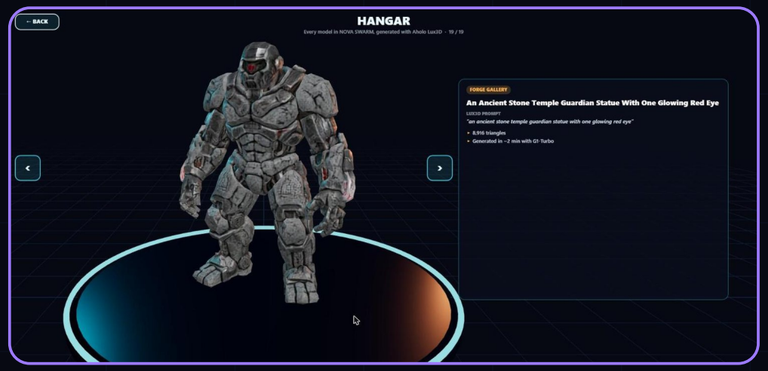
Game feel, built on the generated models



Shatter death



Rail Lance: a piercing beam



Hangar: the prompt behind every model

- **Shatter deaths:** a dying enemy's own Lux3D mesh and textures are cut into chunks that fly out, spin and shrink. It works for every model, including a boss you forged.
- **Boss intro:** letterbox bars, a camera swoop and the boss's name on screen. Players are invulnerable for 2.6 s, so the cinematic never costs a hit.
- **Dash, Chain Lightning and Rail Lance:** a dodge with a cooldown, lightning that arcs between enemies, and a beam through a line of them.
- **Hangar:** a turntable of all 19 models with the exact prompt, triangle count, rigging and clips for each.

ENGINEERING

Quality: a bot that plays the game

A bot plays the real simulation (kites, focuses the boss, dashes, picks random upgrades). It is a crude stand-in for a player, so its numbers compare settings against each other rather than claim absolute truth.

Tuning	Median survival	Beat Warden	Beat capital
First tuning (16 runs)	2:50	1 of 16	0 of 16
After balancing (60 runs)	8:45	42 of 60	20 of 60

Does each feature earn its place? (60 runs each)	Median survival	Beat Warden	Beat capital
Everything on	8:26	39 of 60	10 of 60
Without the two new weapons	7:31	42 of 60	12 of 60
Without dash	5:54	33 of 60	12 of 60

Second table measured mid-tuning. Dash is worth about 2.5 minutes of survival; the new weapons about 1 minute and 35% more kills.

200 tests cover the simulation, network snapshots, host/client sessions, Forge and the relay. A headless two-browser smoke test runs real co-op through a real relay.

One "flaky" test turned out to be a real bug in the test itself: it searched a log for the text "f1", which random channel ids contain about one run in ten. I found the cause instead of retrying, and every commit is now gated on the real test result.

WHAT THE BOT FOUND

- **A crash from the original game:** a dying splitter's child could take its own freed slot and wipe its split setting, so the second child threw. Fixed, with a regression test.
- **83% of damage** was the Warden's five-bullet fan. Now three bullets, slower; same for the capital ship.
- Boss HP was scaled twice (about 2,800 at 2:00). Now gentle.
- Orbit followed total run time, so late launches met a late-game fleet. It now follows time in orbit.

WRAP-UP

Honest scope, and what's next

EXISTED BEFORE

- The simulation, rendering and audio engine
- One CC0 robot used for every character
- 3 weapons, 15 upgrades, 5 enemies, a boss
- Co-op relay and snapshots; Nexconn chat code

BUILT FOR THIS HACKATHON

- The Lux3D pipeline and all 19 models
- Surface, launch, orbit, descent loop
- Forge, gallery fallback, co-op sync
- Part split, shatter, boss intro, Hangar
- Dash and two new weapons; the balance bot

LIMITS, STATED PLAINLY

- Live Forge needs credits; the gallery covers judging
- Dash has no touch button yet
- The free relay sleeps after idle (about 1 min to wake)
- Balance numbers come from a bot, not playtesters
- Rooms are web-only until the Android app is updated

Next: mobile releases (and a touch dash button), image-to-3D companions from a photo, material transfer to re-skin enemies per sector, and more sectors with generated props.

TRY IT IN 60 SECONDS: aholo.vercel.app

1 HANGAR	See all 19 Lux3D models with the prompts that made them.
2 FORGE	Type a boss. Live it takes a few minutes; if busy, a pre-forged one steps in.
3 PLAY SOLO	Hold the mouse to fire, Shift or E to dash, press 1 / 2 / 3 on level-up.
4 LAUNCH	Beat the Warden (it arrives at 2:00) to fly into orbit and meet the capital ship.

Play: aholo.vercel.app **Source: github.com/rohanjain1648/AHOLO**

Models generated with Aholo Lux3D. Fallback robot: RobotExpressive.glb by Tomas Laulhe, CC0, used only if a Lux3D model fails to load. Code: MIT.
Built with Claude Code as a pair-engineer. Libraries: three, ws, gltf-transform, meshoptimizer, sharp, vite, vitest, playwright.